



JABchem



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Past Papers Higher Chemistry

2022 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/120)	%	
A	84+	70.0%	34.9%
B	68+	56.7%	24.1%
C	53+	44.2%	19.3%
D	37+	30.1%	12.9%
No award	<37	<30.1%	8.8%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	16.4 /25	55.1 /95	No Assignment in 2022

2022 Higher Chemistry Marking Scheme

MC Qu	Answer	% Correct	Reasoning																		
1	D	74	☒A Boron is a covalent network due to its high melting point ☒B Neon is monatomic in Group 0 due to its full pouter shell ☒C Sodium is a metal and contains metallic bonding ☒D Sulphur has a covalent S₈ structure and has LdF between molecules																		
2	A	67	☒A Forming a 2+ ion creates a full outer shell and a low 2nd ionisation energy ☒B Forming a 3+ ion creates a full outer shell and a low 3rd ionisation energy ☒C ionisation energy removes electrons and forms positive ions ☒D ionisation energy removes electrons and forms positive ions																		
3	C	48	☒A Intermolecular forces decide the boiling point not the covalent bonds inside ☒B Intermolecular forces decide the boiling point not the covalent bonds inside ☒C Permanent dipole to permanent dipole attractions between polar covalent HCl molecules are stronger than London dispersion forces between H₂ molecules ☒D Van der Waals' forces are never stronger than covalent bonds																		
4	B	63	<table><tr><th>Agent</th><th>Reducing Agent</th><th>Oxidising Agent</th></tr><tr><td>Action of Agent on Another Species</td><td>reduces another species</td><td>oxidises another species</td></tr><tr><td>Action on Agent Itself</td><td>agent is oxidised</td><td>agent is reduced</td></tr><tr><td>Effect on Electrons in Agent</td><td>loss of electrons</td><td>gain of electrons</td></tr><tr><td>Likely Electronegativity of Agent</td><td>low</td><td>high</td></tr><tr><td>Position in Electrochemical series</td><td>top right</td><td>bottom Left</td></tr></table>	Agent	Reducing Agent	Oxidising Agent	Action of Agent on Another Species	reduces another species	oxidises another species	Action on Agent Itself	agent is oxidised	agent is reduced	Effect on Electrons in Agent	loss of electrons	gain of electrons	Likely Electronegativity of Agent	low	high	Position in Electrochemical series	top right	bottom Left
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5	D	45	Redox: $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{Fe}^{2+} \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} + 6\text{Fe}^{3+}$ Oxidation: $6\text{Fe}^{2+} \longrightarrow 6\text{e}^- + 6\text{Fe}^{3+}$ Reduction: $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$																		
6	A	49	<table><tr><td>Formula:</td><td>MgBr₂</td><td>MgSO₄</td></tr><tr><td>Mole ratio:</td><td>1mol : 2mol</td><td>1mol : 1mol</td></tr><tr><td>4mol Br⁻ ions</td><td>2mol : 4mol</td><td></td></tr><tr><td>3mol Mg²⁺ ions</td><td>2mol</td><td>1mol</td></tr><tr><td>1mol SO₄²⁻ ions</td><td></td><td>1mol : 1mol</td></tr></table>	Formula:	MgBr ₂	MgSO ₄	Mole ratio:	1mol : 2mol	1mol : 1mol	4mol Br ⁻ ions	2mol : 4mol		3mol Mg ²⁺ ions	2mol	1mol	1mol SO ₄ ²⁻ ions		1mol : 1mol			
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7	D	84	H H H H O H H H																		

10	C	73	☒ A C=C double bond in prenol molecule would decolourise bromine solution quickly ☒ B C=C double bond in prenol molecule would decolourise bromine solution quickly ☒ C Prenol would decolourise bromine solution and react with hot copper (II) oxide ☒ D Prenol is a primary alcohol and would react with hot copper (II) oxide																																			
11	B	48	Palm oil has iodine number of 48 ∴ 48g of Iodine reacts with 100g of palm oil Olive oil has iodine number of 84 ∴ 84g of Iodine reacts with 100g of olive oil - palm oil must contain less C=C double bonds than olive oil as less iodine is required by palm oil to saturate the molecules completely. - palm oil must be more saturated than olive oil if it contains less C=C bonds - more saturated palm oil molecules fit together better would meaning palm oil molecules are closer together and raises melting point of palm oil.																																			
12	B	86	☒ A head section is polar making it hydrophilic. ☒ B hydrophilic head dissolves in water and hydrophobic tail dissolves in oil ☒ C head section is polar would dissolve in water making it hydrophilic. ☒ D head section is polar would dissolve in water.																																			
13	D	87	<table><tr><td>2-methylbutan-1-ol</td><td>2-methylbutan-2-ol</td><td>butan-1-ol</td><td>butan-2-ol</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td>Primary Alcohol 1 carbon directly attached to the carbon with the -OH group</td><td>Tertiary Alcohol 3 carbons directly attached to the carbon with the -OH group</td><td>Primary Alcohol 1 carbon directly attached to the carbon with the -OH group</td><td>Secondary Alcohol 2 carbons directly attached to the carbon with the -OH group</td></tr></table>	2-methylbutan-1-ol	2-methylbutan-2-ol	butan-1-ol	butan-2-ol					Primary Alcohol 1 carbon directly attached to the carbon with the -OH group	Tertiary Alcohol 3 carbons directly attached to the carbon with the -OH group	Primary Alcohol 1 carbon directly attached to the carbon with the -OH group	Secondary Alcohol 2 carbons directly attached to the carbon with the -OH group																							
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14	C	77	<table><tr><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>cucumber flavour</td><td>vanilla flavour</td><td>ginger flavour</td><td>Orange flavour</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td>mainly non-polar molecule (C=O carbonyl molecule only slightly polar) ∴ oil soluble</td><td>Polar Molecule due to -OH group ∴ water soluble</td><td>Polar Molecule due to 3x -OH groups ∴ water soluble</td><td>Non-polar molecule ∴ oil soluble</td></tr></table>	A	B	C	D	cucumber flavour	vanilla flavour	ginger flavour	Orange flavour					mainly non-polar molecule (C=O carbonyl molecule only slightly polar) ∴ oil soluble	Polar Molecule due to -OH group ∴ water soluble	Polar Molecule due to 3x -OH groups ∴ water soluble	Non-polar molecule ∴ oil soluble																			
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15	B	76	☒ A Carbonyl group would be numbered C ₂ to give functional group lowest numbering system ☒ B Secondary alcohol 4-methylpentan-2-ol oxidises to form the ketone 4-methylpentan-2-one ☒ C Molecule is secondary alcohol and would oxidise to form a ketone not aldehyde ☒ D Molecule is secondary alcohol and would oxidise to form a ketone not aldehyde																																			
16	C	54	☒ A the bottom of the meniscus should be used to measure the volumes in a burette ☒ B rinsing the burette with deionised water will result in the dilution of the next solution in burette ☒ C small volume of the reactant in the burette should be used to rinse the burette before use, the bottom of the meniscus should be used to measure volumes and draining a small volume of acid will remove any air bubble below the tap in the burette. ☒ D the bottom of the meniscus should be used to measure the volumes in a burette																																			
17	B	58	☒ A polar ethanol would not be a solvent to dissolve non-polar lycopene & beta-carotene ☒ B pentane is non-polar and would be a good solvent for non-polar lycopene & beta-carotene ☒ C polar propanoic acid would not be a solvent to dissolve non-polar lycopene & beta-carotene ☒ D polar water would not be a solvent to dissolve non-polar lycopene & beta-carotene																																			
18	D	86	$\text{rate} = \frac{1}{\text{time}} = \frac{1}{0.004} = 250\text{s}$																																			
19	A	63	<table><tr><td>$\text{C}_4\text{H}_{8(g)}$</td><td>+</td><td>$6\text{O}_{2(g)}$</td><td>→</td><td>$4\text{CO}_{2(g)}$</td><td>+</td><td>$4\text{H}_2\text{O}_{(g)}$</td></tr><tr><td>1mol</td><td></td><td>6mol</td><td></td><td>4mol</td><td></td><td>4mol</td></tr><tr><td>1vol</td><td></td><td>6vol</td><td></td><td>4vol</td><td></td><td>4vol</td></tr><tr><td>100cm³</td><td></td><td>600cm³</td><td></td><td>400cm³</td><td></td><td>400cm³</td></tr><tr><td colspan="3">700cm³ used up</td><td></td><td colspan="3">800cm³ produced</td></tr></table>	$\text{C}_4\text{H}_{8(g)}$	+	$6\text{O}_{2(g)}$	→	$4\text{CO}_{2(g)}$	+	$4\text{H}_2\text{O}_{(g)}$	1mol		6mol		4mol		4mol	1vol		6vol		4vol		4vol	100cm ³		600cm ³		400cm ³		400cm ³	700cm ³ used up				800cm ³ produced		
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20	C	58	☒ A No effect as neither Na^+ or Cl^- ions is a reactant or product and don't react with a reactant/product ☒ B H^+ ions in $\text{HCl}_{(\text{aq})}$ increases concentration of a product $\therefore$ equilibrium shifts to left ☒ C OH^- ions in $\text{NaOH}_{(\text{aq})}$ neutralises H^+ in products $\therefore$ equilibrium shifts to right to replace H^+ ions ☒ D CH_3COO^- ions in $\text{CH}_3\text{COONa}_{(\text{aq})}$ increases concentration of product $\therefore$ equilibrium shifts to left																								
21	D	75	<table><tr><th>Quantity</th><th>Measured</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>Enthalpy of Reactants (kJ mol^{-1})</td><td>Where R starts on y-axis</td><td>30</td><td>30</td><td>30</td><td>30</td></tr><tr><td>Activation Energy of Forward Reaction (kJ mol^{-1})</td><td>Difference between R and top of hill</td><td>$80-30 = 50$</td><td>$110-30 = 80$</td><td>$110-30 = 80$</td><td>$140-30 = 110$</td></tr><tr><td>Activation Energy of Reverse Reaction (kJ mol^{-1})</td><td>Difference between P and top of hill</td><td>$80-40 = 40$</td><td>$110-40 = 70$</td><td>$110-70 = 40$</td><td>$140-70 = 70$</td></tr></table>	Quantity	Measured	A	B	C	D	Enthalpy of Reactants (kJ mol^{-1})	Where R starts on y-axis	30	30	30	30	Activation Energy of Forward Reaction (kJ mol^{-1})	Difference between R and top of hill	$80-30 = 50$	$110-30 = 80$	$110-30 = 80$	$140-30 = 110$	Activation Energy of Reverse Reaction (kJ mol^{-1})	Difference between P and top of hill	$80-40 = 40$	$110-40 = 70$	$110-70 = 40$	$140-70 = 70$
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22	B	78	$\Delta H_1 = \Delta H_2 + \Delta H_3 + \Delta H_4$$\Delta H_4 = \Delta H_1 - \Delta H_2 - \Delta H_3$$\Delta H_4 = -210 - (-50) - (-86)$$\Delta H_4 = -74 \text{ kJ mol}^{-1}$ But ΔH for Z to Y = $+74 \text{ kJ mol}^{-1}$																								
23	A	62	50 cm^3 diluted in a 250 cm^3 standard/volumetric flask gives 1 in 5 dilution. 0.100 mol l^{-1} given 1 in 5 dilution results in solution becoming 0.02 mol l^{-1} (or $2.0 \times 10^{-2} \text{ mol l}^{-1}$)																								
24	B	48	☒ A gfm $\text{AgF} = 107.9 \therefore n = m/\text{gfm} = 2.868/107.9 = 0.0266 \text{ mol}$ ☒ B gfm $\text{AgCl} = 143.4 \therefore n = m/\text{gfm} = 2.868/143.4 = 0.0200 \text{ mol}$ ☒ C gfm $\text{AgBr} = 187.8 \therefore n = m/\text{gfm} = 2.868/187.8 = 0.0153 \text{ mol}$ ☒ D gfm $\text{AgI} = 234.8 \therefore n = m/\text{gfm} = 2.868/234.8 = 0.0122 \text{ mol}$																								
25	A	47	☒ A 10 cm^3 of water is better measured in a measuring cylinder and titration carried out in conical flask ☒ B beakers are not as accurate as measuring cylinders for measuring volumes ☒ C Volumetric/standard flasks are used to make up solutions of accurately known concentration ☒ D Volumetric/standard flasks are used to make up solutions of accurately known concentration																								

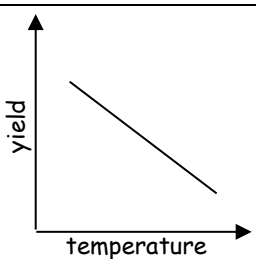
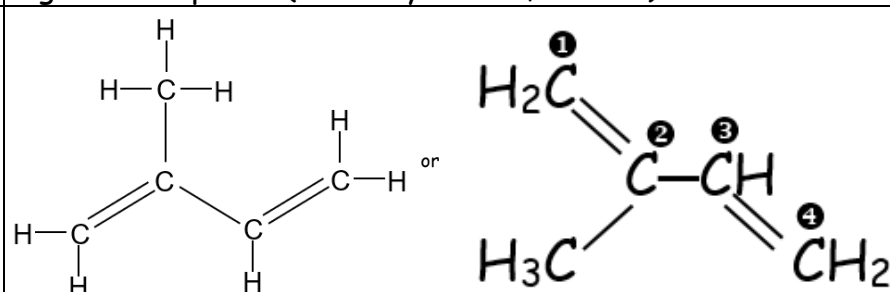
2022 Higher Chemistry Marking Scheme

Long Qu	Answer	Reasoning		
1a(i)	one answer from:	atoms/nuclei have the same attraction for the bonding electrons	same electronegativity/ electronegativity values given	Bonding electrons shared equally (between the atoms).
1a(ii)	greater nuclear charge	Across a period, the number of protons increases giving a greater nuclear charge. The greater nuclear charge pulls the outer electron shell further which reduces the size of the atom.		
1b(i)	Answer to include:	The energy required to remove 1 mole of electrons from one mole of atoms in the gaseous state.		
1b(ii)	One answer from:	More shells so increased screening/shielding covalent radius increases atom size increases more shells } so attraction of { nucleus protons } for outer electrons decreases		
1c(i)	Answer to include:	Hydrogen bonding (1 mark) 1 mark for either: Hydrogen bonding occurs between hydrogen bonded to N, O or F (all 3 elements needed) The attraction between δ+ end on a permanent dipole is strongly attracted to the δ- end of a neighbouring permanent dipole in molecules with hydrogen and a atom with high electronegativity		
1c(ii)	Answer to include:	1 st Mark:	London dispersion forces become stronger (moving from HCl to HI)	
		2 nd Mark:	Number of electrons increases (moving from HCl to HI)	
2a	$3\text{KClO}_4 + 8\text{Al}$ $\downarrow$ $3\text{KCl} + 4\text{Al}_2\text{O}_3$	$3\text{KClO}_4 + 8\text{Al} \longrightarrow 3\text{KCl} + 4\text{Al}_2\text{O}_3$		
2b(i)	1.35	$\text{gfm KClO}_4 = 122.6\text{g}$ $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{4.6}{122.6} = 0.0375\text{mol}$ $2\text{KClO}_3 \longrightarrow 3\text{O}_2 + 2\text{KCl}$ $\begin{array}{ccc} 2\text{mol} & & 3\text{mol} \\ 0.0375\text{mol} & & 0.0563\text{mol} \end{array}$ $\text{Volume} = \text{no. of mol} \times \text{Molar Volume} = 0.0563\text{mol} \times 24\text{litres mol}^{-1} = 1.35\text{litres}$		
2b(ii)	no effect	Catalysts speed up chemical reaction but do not get used up in that chemical reaction. The enthalpy change is the same for the catalysed and the non-catalysed route due to Hess's Law.		
2b(iii)	2595.6	$5.5\text{g} \longleftrightarrow 103\text{kJ}$ $1\text{mol} = 138.6\text{g} \longleftrightarrow 103\text{kJ} \times \frac{138.6}{5.5}$ $= 2595.6\text{kJ}$		
2b(iv)	Answer to include:	1 st Mark:	Increases the number of particles with energy 2 equal to or greater than the activation energy	
		2 nd Mark:	Increases the number of particles with (sufficient) energy to form an activated complex/to react	
2c	Sodium	Peak B at 590nm. Sodium gives a flame colour at 589nm.		
3	Open Ended Question:	3 mark answer		
		2 mark answer		
		1 mark answer		
		Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.		
		Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.		
		Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.		

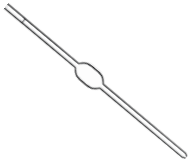
4a(i)	Ester link	—O—H hydroxyl group	$\text{—}\overset{\text{O}}{\parallel}\text{C—OH}$ carboxyl group	$\text{—}\overset{\text{O}}{\parallel}\text{C—O—}$ ester link	$\text{—}\overset{\text{O}}{\parallel}\text{C—}$ carbonyl group																																								
4a(ii)	Diagram showing:																																												
4a(iii)	Structure of Heptan-1-ol or other C ₇ alcohol listed:	Propyl octanoate has a total of 11 carbons. For the isomer of propyl octanoate to be an ester with butanoic acid being released during hydrolysis, the alcohol released by this hydrolysis must have seven carbons. There are 38 possible seven carbon alcohols. (I think!) <table><tr><td>heptan-1-ol</td><td>heptan-2-ol</td><td>heptan-3-ol</td><td>heptan-4-ol</td></tr><tr><td>2-methylhexan-1-ol</td><td>3-methylhexan-1-ol</td><td>4-methylhexan-1-ol</td><td>5-methylhexan-1-ol</td></tr><tr><td>2-methylhexan-2-ol</td><td>3-methylhexan-2-ol</td><td>4-methylhexan-2-ol</td><td>5-methylhexan-2-ol</td></tr><tr><td>2-methylhexan-3-ol</td><td>3-methylhexan-3-ol</td><td>4-methylhexan-3-ol</td><td>5-methylhexan-3-ol</td></tr><tr><td>2,2-dimethylpentan-1-ol</td><td>2,3-dimethylpentan-1-ol</td><td>2,4-dimethylpentan-1-ol</td><td>3,3-dimethylpentan-1-ol</td></tr><tr><td>3,4-dimethylpentan-1-ol</td><td>4,4-dimethylpentan-1-ol</td><td>2,3-dimethylpentan-2-ol</td><td>2,4-dimethylpentan-2-ol</td></tr><tr><td>3,3-dimethylpentan-2-ol</td><td>3,4-dimethylpentan-2-ol</td><td>4,4-dimethylpentan-2-ol</td><td>2,2-dimethylpentan-3-ol</td></tr><tr><td>2,3-dimethylpentan-3-ol</td><td>2,4-dimethylpentan-3-ol</td><td>2,2,3-trimethylbutan-1-ol</td><td>2,3,3-trimethylbutan-1-ol</td></tr><tr><td>2,3,3-trimethylbutan-2-ol</td><td>3-ethylpentan-1-ol</td><td>3-ethylpentan-2-ol</td><td>3-ethylpentan-3-ol</td></tr><tr><td>2-ethyl-2-methylbutan-1-ol</td><td>2-ethyl-3-methylbutan-1-ol</td><td></td><td></td></tr></table>				heptan-1-ol	heptan-2-ol	heptan-3-ol	heptan-4-ol	2-methylhexan-1-ol	3-methylhexan-1-ol	4-methylhexan-1-ol	5-methylhexan-1-ol	2-methylhexan-2-ol	3-methylhexan-2-ol	4-methylhexan-2-ol	5-methylhexan-2-ol	2-methylhexan-3-ol	3-methylhexan-3-ol	4-methylhexan-3-ol	5-methylhexan-3-ol	2,2-dimethylpentan-1-ol	2,3-dimethylpentan-1-ol	2,4-dimethylpentan-1-ol	3,3-dimethylpentan-1-ol	3,4-dimethylpentan-1-ol	4,4-dimethylpentan-1-ol	2,3-dimethylpentan-2-ol	2,4-dimethylpentan-2-ol	3,3-dimethylpentan-2-ol	3,4-dimethylpentan-2-ol	4,4-dimethylpentan-2-ol	2,2-dimethylpentan-3-ol	2,3-dimethylpentan-3-ol	2,4-dimethylpentan-3-ol	2,2,3-trimethylbutan-1-ol	2,3,3-trimethylbutan-1-ol	2,3,3-trimethylbutan-2-ol	3-ethylpentan-1-ol	3-ethylpentan-2-ol	3-ethylpentan-3-ol	2-ethyl-2-methylbutan-1-ol	2-ethyl-3-methylbutan-1-ol		
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4b(i)	35-45	<table><tr><td>Triglyceride</td><td>Glyceryl trilinoleate</td><td>Glyceryl tricaprates</td><td>Difference</td></tr><tr><td>Number of Carbons</td><td>57 carbons</td><td>33 carbons</td><td>24 carbons</td></tr><tr><td>Absorbance Units</td><td>19.3</td><td>16.1</td><td>3.2</td></tr></table> 3.2 difference in absorbance units = 24 carbons Glyceryl trilaurate = 17.5 absorbance units (1.4 units above Glyceryl tricaprates) 1.4 difference in absorbance units = $24 \times 1.4 / 3.2 = 10.5$ carbons ∴ Estimate of number of carbons in glyceryl trilaurate = 24 + 11 = 35				Triglyceride	Glyceryl trilinoleate	Glyceryl tricaprates	Difference	Number of Carbons	57 carbons	33 carbons	24 carbons	Absorbance Units	19.3	16.1	3.2																												
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4b(ii)	glyceryl trilinoleate	The lower the melting point, the higher the number of C=C double bonds in molecule. Oil molecules do not fit as close together due to the change of direction in the carbon chain after a C=C double bond. The further apart the molecules are, the lower the melting point as less energy is needed to separate the molecules into a liquid as there are weaker van der Waals' between oil molecules.																																											
4c(i)	by react with glycerol	Fatty acids from edible oils <u>react</u> with glycerol by condensation reaction. One or two fatty acids react with glycerol to form an emulsifier. This will leave at least one polar -OH group on the glycerol part of the molecule needed to form the hydrophilic head on the emulsifier.																																											
4c(ii)	Answer to include:	<table><tr><td>1st Mark:</td><td colspan="3">Correctly identifying that the 2 emulsifier has two parts with different polarities or two parts that are hydrophobic/hydrophilic.</td></tr><tr><td>2nd Mark:</td><td> Hydrophobic part hydrocarbon chain fatty acid chain non-polar part </td><td> dissolves in non-polar liquids </td><td> hydrophilic part hydroxyl groups polar part </td><td> dissolves in polar liquids </td></tr></table>				1 st Mark:	Correctly identifying that the 2 emulsifier has two parts with different polarities or two parts that are hydrophobic/hydrophilic.			2 nd Mark:	Hydrophobic part hydrocarbon chain fatty acid chain non-polar part	dissolves in non-polar liquids	hydrophilic part hydroxyl groups polar part	dissolves in polar liquids																															
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5a	3-methylbutan-1-ol	H H</																																											

5b(ii)	To provide H ⁺ ions	H ⁺ ions are a reactant on the left hand side of the equation. If the reactants are not acidified then one of the reactants will be absent and the chemical reaction will not proceed.																					
5b(iii)	orange → green	<table><tr><th>Oxidising Agent</th><th>Start Colour</th><th>End Colour</th></tr><tr><td>Acidified Dichromate</td><td>Orange</td><td>Green</td></tr><tr><td>Fehling's Solution</td><td>Blue</td><td>Brick Red (orange)</td></tr><tr><td>Hot copper (II) oxide</td><td>Black</td><td>Brown</td></tr><tr><td>Tollen's Reagent</td><td>(Colourless)</td><td>Silver mirror produced</td></tr></table>					Oxidising Agent	Start Colour	End Colour	Acidified Dichromate	Orange	Green	Fehling's Solution	Blue	Brick Red (orange)	Hot copper (II) oxide	Black	Brown	Tollen's Reagent	(Colourless)	Silver mirror produced		
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5b(iv)	Tollen's Reagent	<table><tr><th>Oxidising Agent</th><th>Reactant(s)</th><th>Product(s)</th></tr><tr><td>Acidified Dichromate</td><td>Cr₂O₇²⁻ + 14H⁺ + 6e⁻</td><td>→ 2Cr³⁺ + 7H₂O</td></tr><tr><td>Fehling's Solution</td><td>Cu²⁺ + e⁻</td><td>→ Cu⁺</td></tr><tr><td>Hot copper (II) oxide</td><td>Cu²⁺ + 2e⁻</td><td>→ Cu</td></tr><tr><td>Tollen's Reagent</td><td>Ag⁺ + e⁻</td><td>→ Ag</td></tr></table>					Oxidising Agent	Reactant(s)	Product(s)	Acidified Dichromate	Cr ₂ O ₇ ²⁻ + 14H ⁺ + 6e ⁻	→ 2Cr ³⁺ + 7H ₂ O	Fehling's Solution	Cu ²⁺ + e ⁻	→ Cu ⁺	Hot copper (II) oxide	Cu ²⁺ + 2e ⁻	→ Cu	Tollen's Reagent	Ag ⁺ + e ⁻	→ Ag		
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5b(v)	tertiary alcohols (do not oxidise)	<table><tr><td rowspan="3">Oxidation of Alcohols</td><td>Primary alcohol</td><td>→</td><td>Aldehyde</td><td>→</td><td>Carboxylic acid</td></tr><tr><td>Secondary alcohol</td><td>→</td><td>Ketone</td><td>→</td><td>[No oxidation]</td></tr><tr><td>Tertiary alcohol</td><td>→</td><td>[No oxidation]</td><td></td><td></td></tr></table>					Oxidation of Alcohols	Primary alcohol	→	Aldehyde	→	Carboxylic acid	Secondary alcohol	→	Ketone	→	[No oxidation]	Tertiary alcohol	→	[No oxidation]			
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5b(vi)	<table><tr><td>1 : 10</td></tr><tr><td>1 : 8</td></tr></table>	1 : 10	1 : 8	<table><tr><th>Chemical</th><th>Formula</th><th>No. of O</th><th>No. of H</th><th>Oxygen : Hydrogen ratio</th></tr><tr><td>butan-1-ol</td><td>C₄H₉OH</td><td>1</td><td>10</td><td>1:10</td></tr><tr><td>butanal</td><td>C₄H₈O</td><td>1</td><td>8</td><td>1:8</td></tr></table>					Chemical	Formula	No. of O	No. of H	Oxygen : Hydrogen ratio	butan-1-ol	C ₄ H ₉ OH	1	10	1:10	butanal	C ₄ H ₈ O	1	8	1:8
1 : 10																							
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butan-1-ol	C ₄ H ₉ OH	1	10	1:10																			
butanal	C ₄ H ₈ O	1	8	1:8																			
6a(i)	biological catalyst	An enzyme is a specially-shaped protein which acts as a biological catalyst, catalysing chemical reactions in the body at 37°C.																					
6a(ii)A	one peptide link circled: 																						
6a(ii)B	one amino acid structure from:																						
6a(ii)C	amino acid which must be obtained through diet	Essential amino acids are amino acids which must be obtained from your diet for a healthy diet to be obtained. These amino acids cannot be made by the body.																					
6a(ii)D	condensation	A condensation reaction occurs when two molecules join together to form a bigger molecule and water is removed at the join. Other small molecules can also be removed instead of water.																					
6a(iii)	Answer to include:	<table><tr><td>1st Mark:</td><td>enzyme becomes denatured/enzyme changes shape</td></tr><tr><td>2nd Mark:</td><td>Intermolecular/hydrogen bonds are broken</td></tr></table>					1 st Mark:	enzyme becomes denatured/enzyme changes shape	2 nd Mark:	Intermolecular/hydrogen bonds are broken													
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2 nd Mark:	Intermolecular/hydrogen bonds are broken																						
6a(iv)	Answer to include one of:																						

6b(i)	one answer from:	To prevent unwanted oxidation	to oxidise in place of the compounds they have been added to protect	to stop (oxidation of edible oils) food acquiring a rancid flavour.																
6b(ii)	answer to include:	1 st Mark: Vitamin C molecule is polar due to its hydroxyl groups	or	Vitamin C can form hydrogen bonds due to its hydroxyl groups																
		2 nd Mark: Vitamin C is soluble in water because of interactions of polar -OH groups in Vitamin C with polar -OH groups in water.																		
6c	975g 2 marks for mass 1 mark for units	1kg body weight ↔ 3mg solanine 65kg body weight ↔ 3mg solanine × ⁶⁵ / ₁ =195g solanine 0.2mg solanine ↔ 1g of potato 195mg solanine ↔ 1g of potato × ¹⁹⁵ / _{0.2} = 975g of potato																		
7a	0.203g	Heat Energy = Specific Heat Capacity × Mass × Change In Temperature E _h = c × m × ΔT E _h = 4.18 kJ kg ⁻¹ °C ⁻¹ × 0.1kg × 27°C E _h = 11.3 kJ gfm Heptane CH ₄ = (1×12) + (4×1) = 12 + 4 = 16g 1 mol CH ₄ = 891 kJ ↔ 16g 11.3 kJ ↔ 16g × ^{11.3} / ₈₉₁ = 0.203g																		
7b	-816	<table><tr><th colspan="2">Bond Breaking Steps (endothermic)</th><th colspan="2">Bond Forming Steps (exothermic)</th></tr><tr><td>4×C-H bonds</td><td>4× 412kJ = 1648kJ</td><td>2×C=O bonds</td><td>2× 804kJ = 1608kJ</td></tr><tr><td>2×O=O bond</td><td>2× 498kJ = 996kJ</td><td>4×O-H bonds</td><td>4× 463kJ = 1852kJ</td></tr><tr><td>Total bond breaking</td><td>= 2644kJ</td><td>Total bond Forming</td><td>= 3460kJ</td></tr></table> Enthalpy change = ΣBond Breaking Steps - ΣBond forming steps = 2644 - 3460 = -816kJ mol ⁻¹			Bond Breaking Steps (endothermic)		Bond Forming Steps (exothermic)		4×C-H bonds	4× 412kJ = 1648kJ	2×C=O bonds	2× 804kJ = 1608kJ	2×O=O bond	2× 498kJ = 996kJ	4×O-H bonds	4× 463kJ = 1852kJ	Total bond breaking	= 2644kJ	Total bond Forming	= 3460kJ
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7c	17.6%	atom economy = $\frac{\text{mass of useful products}}{\text{total mass of reactants}} \times 100 = \frac{(3 \times 2)}{(1 \times 16) + (1 \times 18)} \times 100 = 17.6\%$																		
7d	High Low	<table><tr><th>Change in Temperature</th><th>Change In Pressure</th></tr><tr><td>Maximising Yield of NO₂ = more reverse reaction - reverse reaction is endothermic - Increasing temperature favours the endothermic reaction - HIGH temperature increases reverse reaction - HIGH temperature increases yield of NO₂ </td><td>Maximising Yield of NO₂ = more reverse reaction - reverse reaction increases pressure (1vol→2vol) - Decreasing pressure favours the pressure-increasing reaction - LOW pressure increases reverse reaction - LOW pressure increases yield of NO₂ </td></tr></table>			Change in Temperature	Change In Pressure	Maximising Yield of NO ₂ = more reverse reaction - reverse reaction is endothermic - Increasing temperature favours the endothermic reaction - HIGH temperature increases reverse reaction - HIGH temperature increases yield of NO₂	Maximising Yield of NO ₂ = more reverse reaction - reverse reaction increases pressure (1vol→2vol) - Decreasing pressure favours the pressure-increasing reaction - LOW pressure increases reverse reaction - LOW pressure increases yield of NO₂												
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7e(i)	-4632	① C_(s) + 3H_{2(g)} + N_{2(g)} → CH₃NHNH_{2(l)} ΔH=+54 kJ mol⁻¹ ② N_{2(g)} + 2O_{2(g)} → N₂O_{4(g)} ΔH=-20 kJ mol⁻¹ ③ C_(s) + O_{2(g)} → CO_{2(g)} ΔH=-394 kJ mol⁻¹ ④ H_{2(g)} + $\frac{1}{2}$O_{2(g)} → H₂O_(l) ΔH=-286 kJ mol⁻¹ ⑤ H₂O_(l) → H₂O_(g) ΔH=+41 kJ mol⁻¹ ①×-4 4CH₃NHNH_{2(l)} → 4C_(s) + 12H_{2(g)} + 4N_{2(g)} ΔH=-216 kJ mol⁻¹ ②×-5 5N₂O_{4(g)} → 5N_{2(g)} + 10O_{2(g)} ΔH=+100 kJ mol⁻¹ ③×4 4C_(s) + 4O_{2(g)} → 4CO_{2(g)} ΔH=-1576 kJ mol⁻¹ ④×12 12H_{2(g)} + 6O_{2(g)} → 12H₂O_(l) ΔH=-3432 kJ mol⁻¹ ⑤×12 12H₂O_(l) → 12H₂O_(g) ΔH=+492 kJ mol⁻¹ add 5N₂O_{4(g)} + 4CH₃NHNH_{2(l)} → 4CO_{2(g)} + 12H₂O_(g) + 9N_{2(g)} ΔH=-4632 kJ mol⁻¹																		
7e(ii)	H H H H-C-N-N-H H	<table><tr><th>Element</th><th>Valency</th><th>No of Bonds made by element</th></tr><tr><td>C</td><td>4</td><td>4</td></tr><tr><td>N</td><td>3</td><td>3</td></tr><tr><td>H</td><td>1</td><td>1</td></tr></table>			Element	Valency	No of Bonds made by element	C	4	4	N	3	3	H	1	1				
Element	Valency	No of Bonds made by element																		
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8a(i)	increase increase no effect	A catalyst increases the rate of both the forward and reverse reactions by lowering the activation energies of both the forward and reverse reactions. The position of equilibrium is not changed but the time to get to equilibrium is shortened.				
8a(ii)		The forward reaction in the water-gas shift reaction is exothermic. - Increasing the temperature favours the endothermic reaction - Reverse reaction is endothermic - Reverse reaction is favoured by increasing the temperature - Less products formed as temperature increase - Graph has decreasing slope as yield decreases as temperature increases				
8b	Calculation showing:	gfm sorbic acid $C_6H_8O_2 = (6 \times 12) + (8 \times 1) + (2 \times 16) = 72 + 8 + 32 = 112$ $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{7}{112} = 0.0625 \text{ mol (available)}$ n KOH = volume $\times$ concentration = $0.25 \text{ litres} \times 0.5 \text{ mol l}^{-1} = 0.125 \text{ mol}$ $C_6H_8O_2 + KOH \longrightarrow H_2O + C_6H_7O_2$ $\begin{matrix} 1 \text{ mol} \\ 0.125 \text{ mol} \\ \text{(required)} \end{matrix}$ Less no. of mol of sorbic acid available than is required $\therefore$ Sorbic acid is limiting reactant and KOH is in excess				
8c	2.52×10^{-5} or 0.0000252	$1\% = 1 \text{ g per } 100 \text{ cm}^3$ $0.002\% = 0.002 \text{ g per } 100 \text{ cm}^3$ $100 \text{ cm}^3 = 0.002 \text{ g}$ $330 \text{ cm}^3 = 0.002 \text{ g} \times \frac{330}{100} = 0.0066 \text{ g}$ gfm = 261.8g $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{0.0066}{261.8} = 2.52 \times 10^{-5} \text{ mol}$				
8d(i)A	non-water soluble or volatile or aroma	Essential oils are concentrated extracts of volatile, non-water soluble aroma compounds from plants - mixtures of many different compounds. - widely used in <table><tr><td>perfumes</td><td>cosmetic products</td><td>cleaning products</td><td>flavourings in foods</td></tr></table>	perfumes	cosmetic products	cleaning products	flavourings in foods
perfumes	cosmetic products	cleaning products	flavourings in foods			
8d(i)B Part I	terpene	Terpenes are key components in most essential oils. Terpenes are unsaturated compounds formed by joining together isoprene (2-methylbuta-1,3-diene) units.				
8d(i)B Part II	correct structure drawn of 2-methylbut-1,3-diene					
8d(i)B Part III	3	Formula of zingiberene: $C_{15}H_{24}$ Formula of isoprene: C_5H_8 $\therefore$ 3 isoprene units join together				
8d(ii)A	water or H_2O	The difference between the two molecules is the a $C=C$ double bond is formed in the product and an H atom was removed on one side where the $C=C$ double bond formed and a OH group was removed from the other side of where the $C=C$ double bond formed.				

8d(ii)B	Hydroxyl group and Carbonyl Group	—O—H hydroxyl group $\begin{array}{c} \text{O} \\ \\ \text{—C—OH} \end{array}$ carboxyl group $\begin{array}{c} \text{O} \\ \\ \text{—C—} \end{array}$ carbonyl group																		
9	Open Ended Question: <table><tr><th colspan="4">Candidate Marks</th></tr><tr><th>0 marks</th><th>1 mark</th><th>2 marks</th><th>3 marks</th></tr><tr><td>51%</td><td>24%</td><td>16%</td><td>8%</td></tr></table>	Candidate Marks				0 marks	1 mark	2 marks	3 marks	51%	24%	16%	8%	<table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
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10a(i)	One from:																			
10a(ii)	1+5	Refrigerant Compound 1 has 2 carbons, 4 fluorines and 2 bromines Refrigerant Compound 1 has 2 carbons, 4 fluorines and 2 chlorines																		
10a(iii)	Carbon dioxide and ammonia do not contain halogens or Carbon dioxide and ammonia do not damage the ozone layer	CO ₂ and NH ₃ lack group 7 elements (halogen) atoms in their structure. All refrigerant compounds in table have halogen atoms in their structure.																		
10b(i)	Species (atoms/molecules/particles) with unpaired electron	Free radicals are atoms or molecules that are highly reactive due to the presence of an unpaired electron.																		
10b(ii)A	Initiation	<table><tr><th>Step</th><th>Reactants (before Arrow)</th><th>Products (after Arrow)</th></tr><tr><td>Initiation</td><td>No free radicals on Reactant Side</td><td>Free radicals on Product Side</td></tr><tr><td>Propagation</td><td colspan="2">Free Radicals found on both sides of arrow</td></tr><tr><td>Termination</td><td>Free radicals on Reactant Side</td><td>No free radicals on Product Side</td></tr></table>	Step	Reactants (before Arrow)	Products (after Arrow)	Initiation	No free radicals on Reactant Side	Free radicals on Product Side	Propagation	Free Radicals found on both sides of arrow		Termination	Free radicals on Reactant Side	No free radicals on Product Side						
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10b(ii)B	One from:	$\text{CH}_3\text{F} + \text{F}^\bullet \longrightarrow \bullet\text{CH}_2\text{F} + \text{HF}$ $\bullet\text{CH}_2\text{F} + \text{F}_2 \longrightarrow \text{CH}_2\text{F}_2 + \text{F}^\bullet$ $\text{F}_2 + \bullet\text{CH}_3 \longrightarrow \text{CH}_3\text{F} + \text{F}^\bullet$ $\bullet\text{CH}_2\text{F} + \text{HF} \longrightarrow \text{CH}_2\text{F}_2 + \text{H}^\bullet$																		
10c	0.208	0.05kg → 0.025kg difluoromethane → 0.025kg pentafluoroethane $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{25\text{g}}{120 \text{ g mol}^{-1}} = 0.208 \text{ mol}$																		
11a(i)	water and carbon dioxide	<table><tr><td>copper (II) carbonate</td><td>+</td><td>ethanoic acid</td><td>→</td><td>copper (II) ethanoate</td><td>+</td><td>water</td><td>+</td><td>carbon dioxide</td></tr><tr><td>metal carbonate</td><td>+</td><td>acid</td><td>→</td><td>salt</td><td>+</td><td>water</td><td>+</td><td>carbon dioxide</td></tr></table>	copper (II) carbonate	+	ethanoic acid	→	copper (II) ethanoate	+	water	+	carbon dioxide	metal carbonate	+	acid	→	salt	+	water	+	carbon dioxide
copper (II) carbonate	+	ethanoic acid	→	copper (II) ethanoate	+	water	+	carbon dioxide												
metal carbonate	+	acid	→	salt	+	water	+	carbon dioxide												
11a(ii)	Cu ²⁺ (CH ₃ COO ⁻) ₂	Copper (II) has a valency of 2 and forms Cu ²⁺ ions Ethanoate ions has a formula of CH ₃ COO ⁻ and valency of 1. Formula of copper (II) ethanoate is Cu(CH ₃ COO). Ionic formula of copper (II) ethanoate is Cu ²⁺ (CH ₃ COO ⁻) ₂																		

11b	Answer to include:	1 mark	1mark	1mark
		Dissolve oxalic acid (in a small volume of deionised water)	Transfer quantitatively oxalic acid solution to standard/volumetric flask including rinsings/washings	Fill volumetric/standard flask up to mark (with deionised water)
11c(i)		Volumetric bulb pipette to be drawn showing: - volumetric mark/line - end of pipette must narrow to a point A graduated pipette would also be acceptable.		
11c(ii)	pink → colourless	Colour in conical flask at start: <u>pink</u> as sodium hydroxide solution is in conical flask at start and phenolphthalein is pink in alkaline conditions Colour in conical flask at end: <u>colourless</u> as sodium hydroxide in conical flask has been neutralised by the addition of oxalic acid from the burette. Phenolphthalein is colourless in acidic/neutral conditions		
11c(iii)	concordant	Results in a titration are described as concordant when the individual titres are within 0.2cm ³ of each other. This ignores the rough titre and any rogue results.		
11d	0.27	Oxalic acid no. of mol = volume x concentration = 0.02675litres x 0.126mol l ⁻¹ = 0.00337mol $\begin{array}{ccc} \text{H}_2\text{C}_2\text{O}_4 & + & 2\text{NaOH} \longrightarrow \text{Na}_2\text{C}_2\text{O}_4 + 2\text{H}_2\text{O} \\ \text{1mol} & & \text{2mol} \\ \text{0.00337mol} & & \text{0.00674mol} \end{array}$ $\text{concentration} = \frac{\text{no. of mol}}{\text{volume}} = \frac{0.00674_{\text{mol}}}{0.025_{\text{litres}}} = 0.270 \text{ mol l}^{-1}$		

2022 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	D	16	3	7	74	0	0
2	A	67	15	15	2	0	0
3	C	17	30	48	4	1	0
4	B	16	63	11	10	0	0
5	D	17	18	19	45	1	0
6	A	49	21	18	11	1	0
7	D	2	5	9	84	0	0
8	C	16	20	52	11	1	0
9	A	82	5	7	6	0	0
10	C	1	6	73	20	0	0
11	B	10	48	26	15	0	0
12	B	3	86	8	3	0	0
13	D	3	8	2	87	0	0
14	C	12	6	77	5	0	0
15	B	8	76	7	9	0	0
16	C	8	26	54	12	0	0
17	B	14	58	12	15	1	0
18	D	11	1	2	86	0	0
19	A	63	11	22	3	1	0
20	C	6	15	58	19	1	0
21	D	1	9	15	75	1	0
22	B	15	78	5	2	0	0
23	A	62	18	6	14	1	0
24	B	14	48	15	21	2	0
25	A	47	5	39	7	0	1

2022 Higher Chemistry Section 2 Marks Data Analysis

2022 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.71	71%	46	2%
Q1aai	1	0.79	79%	43	1%
Q1bi	1	0.36	36%	39	1%
Q1bii	1	0.54	54%	34	1%
Q1ci	2	0.95	47%	36	1%
Q1cii	2	0.96	48%	51	3%
Q2a	1	0.83	83%	23	1%
Q2bi	2	1.41	71%	58	2%
Q2bii	1	0.70	70%	46	1%
Q2biii	1	0.67	67%	50	3%
Q2biv	2	0.95	47%	50	1%
Q2c	1	0.77	77%	15	1%
Q3	3	1.34	45%	65	4%
Q4ai	1	0.72	72%	47	1%
Q4aai	1	0.73	73%	53	2%
Q4aaii	1	0.31	31%	40	3%
Q4bi	1	0.73	73%	16	4%
Q4bii	1	0.79	79%	33	0%
Q4ci	1	0.32	32%	45	10%
Q4cii	2	0.83	41%	52	10%
Q5a	1	0.69	69%	42	0%
Q5bi	1	0.45	45%	54	9%
Q5bii	1	0.33	33%	48	9%
Q5biii	1	0.68	68%	48	2%
Q5biv	1	0.70	70%	48	6%
Q5bv	1	0.77	77%	50	6%
Q5bvi	1	0.62	62%	55	3%
Q6ai	1	0.81	81%	39	2%
Q6aiiA	1	0.87	87%	25	5%
Q6aiiB	1	0.57	57%	62	4%
Q6aiiC	1	0.88	88%	41	1%
Q6aiiD	1	0.69	69%	53	4%
Q6aiii	2	1.23	62%	57	4%
Q6aiv	3	1.89	63%	46	6%
Q6bi	1	0.40	40%	33	2%
Q6bii	2	1.14	57%	61	2%
Q6c	3	2.01	67%	50	2%
Q7a	3	1.60	53%	66	4%
Q7b	2	1.25	62%	58	3%
Q7c	2	1.66	83%	48	1%
Q7d	1	0.54	54%	30	1%
Q7ei	2	0.92	46%	47	4%
Q7eii	1	0.81	81%	30	1%
Q8ai	2	1.41	71%	44	1%
Q8aai	1	0.44	44%	47	2%
Q8b	2	1.52	76%	54	4%
Q8c	2	0.80	40%	51	8%
Q8diA	1	0.50	50%	44	3%
Q8diBi	1	0.82	82%	47	4%
Q8diBii	1	0.55	55%	50	4%
Q8diBiii	1	0.87	87%	36	2%
Q8diiA	1	0.91	91%	30	1%
Q8diiB	1	0.73	73%	48	1%
Q9	3	0.82	27%	59	16%
Q10ai	1	0.64	64%	27	3%
Q10aai	1	0.73	73%	37	3%
Q10aaii	1	0.38	38%	24	7%
Q10bi	1	0.61	61%	50	4%
Q10biiA	1	0.88	88%	43	2%
Q10biiB	1	0.15	15%	34	7%
Q10c	1	0.37	37%	40	8%
Q11ai	1	0.40	40%	41	5%
Q11aai	1	0.19	19%	38	12%
Q11b	3	1.18	39%	60	14%
Q11ci	1	0.22	22%	18	6%
Q11cii	1	0.52	52%	44	2%
Q11ciii	1	0.73	73%	43	7%
Q11d	3	1.77	59%	59	6%
Paper	95	55.02	58%		

2022 Higher Chemistry Section 2 Frequency of Marks Analysis

2022 Item	0	1	2	3
Q1ai	29	71		
Q1aii	21	79		
Q1bi	64	36		
Q1bii	46	54		
Q1ci	15	75	10	
Q1cii	42	19	39	
Q2a	17	83		
Q2bi	15	29	56	
Q2bii	30	70		
Q2biii	33	67		
Q2biv	29	48	23	
Q2c	23	77		
Q3	19	39	31	11
Q4ai	28	72		
Q4aii	27	73		
Q4aiii	69	31		
Q4bi	27	73		
Q4bii	21	79		
Q4ci	68	32		
Q4cii	43	31	26	
Q5a	31	69		
Q5bi	55	45		
Q5bii	67	33		
Q5biii	32	68		
Q5biv	30	70		
Q5bv	23	77		
Q5bvi	38	62		
Q6ai	19	81		
Q6aiiA	13	87		
Q6aiiB	43	57		
Q6aiiC	12	88		
Q6aiiD	31	69		
Q6aiii	12	54	35	
Q6aiv	14	17	35	34
Q6bi	60	40		
Q6bii	25	36	39	
Q6c	16	17	18	49
Q7a	31	20	7	42
Q7b	21	33	46	
Q7c	11	12	77	
Q7d	46	54		
Q7ei	34	40	26	
Q7eii	19	81		
Q8ai	17	25	58	
Q8aii	56	44		
Q8b	14	21	65	
Q8c	39	43	18	
Q8diA	50	50		
Q8diBi	18	82		
Q8diBii	45	55		
Q8diBiii	13	87		
Q8diiA	9	91		
Q8diiB	27	73		
Q9	51	24	16	8
Q10ai	36	64		
Q10aii	27	73		
Q10aiii	62	38		
Q10bi	39	61		
Q10biiA	12	88		
Q10biiB	85	15		
Q10c	63	37		
Q11ai	60	40		
Q11aii	81	19		
Q11b	47	12	17	24
Q11ci	78	22		
Q11cii	48	52		
Q11ciii	27	73		
Q11d	24	20	12	45